

The United States Bloc of the Monetary-Financial-Fiscal Macromodel: The Estimated Model and Its International Linkages

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Abstract

This note writes down, in the standard notation of the VAR literature, the model the MFFM estimates for the United States and the exact channels through which foreign variables feed into it. The US bloc is a 40-variable Bayesian VAR(3): 37 domestic series plus three trade-weighted foreign aggregates (“stars”) — foreign demand, foreign inflation, and a foreign short rate — that enter as *endogenous* columns with their own equations, not as weakly exogenous regressors. Foreign information reaches the domestic block through three explicit channels: the 37×3 lag blocks on the stars, the innovation covariance between domestic and star residuals, and — at solve time — the conditioning of the star paths on the other 32 coupled economies’ solutions through a linear fixed point $x = Mx + d$ with spectral radius $\rho(M) = 0.947$. All coefficients reported here are posterior means read from the shipped model files.

1 The model in one paragraph

The United States is one of 33 coupled blocs in the MFFM’s global system. Its bloc is a constant-coefficient VAR(3) in $n = 40$ variables estimated on $T = 110$ quarters (1998Q3–2025Q4) under a hierarchical Minnesota prior (Litterman, 1986; Giannone et al., 2015), with the 2020Q2–Q4 macro observations treated as missing data and imputed by a Durbin–Koopman simulation smoother inside the Gibbs sampler (Durbin and Koopman, 2002; Tanner and Wong, 1987). Three of the 40 variables are constructed trade-weighted foreign aggregates whose historical values are rebuilt from the other blocs’ completed panels at every sweep of an outer EM-style iteration, and whose future paths are *pinned* to the other blocs’ solutions whenever a coupled (global) solve is requested. This note states each of those objects exactly.

*International Monetary Fund. This note documents the deployed model exactly as shipped; every equation was checked against the implementation (`pipeline/*.m`, `app/src/lib/compute/*.ts`) and every number recomputed from the shipped data artifacts. It accompanies the full user manual (*A guide to the Monetary-Financial-Fiscal Macromodel*), whose notation it follows.

2 The US VAR

2.1 The system

Let $y_t \in \mathbb{R}^{40}$ collect the US variables of Table 5 in the order shipped. The estimated model is the closed reduced-form VAR(p) with intercept,

$$y_t = c + \sum_{\ell=1}^p B_{\ell} y_{t-\ell} + u_t, \quad u_t \sim \text{i.i.d. } \mathcal{N}(0, \Sigma), \quad p = 3, \quad (1)$$

with $c \in \mathbb{R}^{40}$, $B_{\ell} \in \mathbb{R}^{40 \times 40}$, and $\Sigma \in \mathbb{R}^{40 \times 40}$ symmetric positive definite. The estimation sample is $t = 1998\text{Q3}, \dots, 2025\text{Q4}$ ($T = 110$ rows; the regression conditions on the first $p = 3$).¹

Variables tagged *log* in Table 5 enter in the $100 \ln$ space, $\ell_t = 100 \ln Y_t$, so that the app's displayed quarter-on-quarter annualized growth is $g_t = 4(\ell_t - \ell_{t-1}) = 400 \Delta \ln Y_t$; variables tagged *level* (rates, ratios to GDP, survey indices) enter untransformed. The three foreign stars are stored as *level* columns whose values are themselves already annualized-percent constructs (Section 3).

2.2 The domestic/foreign partition

Partition y_t into the $n_x = 37$ domestic variables and the $n_* = 3$ foreign stars,

$$y_t = \begin{pmatrix} x_t \\ x_t^* \end{pmatrix}, \quad x_t^* = \begin{pmatrix} \text{ROWDEM}_t \\ \text{ROWINF}_t \\ \text{ROWFIN}_t \end{pmatrix} = \begin{pmatrix} \text{foreign demand} \\ \text{foreign inflation} \\ \text{foreign short rate} \end{pmatrix}, \quad (2)$$

where x_t stacks the macro core (columns 0–19), the financial block (20–31, containing three of the four US-originated global commons), the fiscal block (32–35), and the oil-price global common GLB_OIL at column 36, in the shipped order of Table 5. Conformably partitioning (1),

$$\begin{pmatrix} x_t \\ x_t^* \end{pmatrix} = \begin{pmatrix} c_x \\ c_* \end{pmatrix} + \sum_{\ell=1}^3 \begin{pmatrix} B_{\ell}^{xx} & B_{\ell}^{x*} \\ B_{\ell}^{*x} & B_{\ell}^{**} \end{pmatrix} \begin{pmatrix} x_{t-\ell} \\ x_{t-\ell}^* \end{pmatrix} + \begin{pmatrix} u_{x,t} \\ u_{*,t} \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \Sigma_{xx} & \Sigma_{x*} \\ \Sigma_{*x} & \Sigma_{**} \end{pmatrix}. \quad (3)$$

This partition makes the foreign feed-in explicit. Foreign variables reach the US model through three channels:

1. **Lagged transmission** through the 37×3 blocks $B_1^{x*}, B_2^{x*}, B_3^{x*}$: every domestic equation contains nine foreign regressors — the three stars at each of three lags — whose coefficients are estimated jointly with everything else under the prior of Section 4. Section 6 reports the posterior means of selected rows of these blocks.

¹Sample and dimensions from the shipped `app/public/data/bvar/US/spec.json` ($n = 40$), `companion_mean.json` ($\hat{\beta}$: 121×40 , $\hat{\Sigma}$: 40×40 , `lags`: 3) and `coverage_manifest.json` (estimation sample 1998Q3–2025Q4, 110 periods). The deployed *display* origin is one quarter later (2026Q1): the production data shift appended one observed quarter and re-anchored the forecast origin without re-estimating the companion, so forecasts shown in the tool run 2026Q2–2031Q1 ($H = 20$).

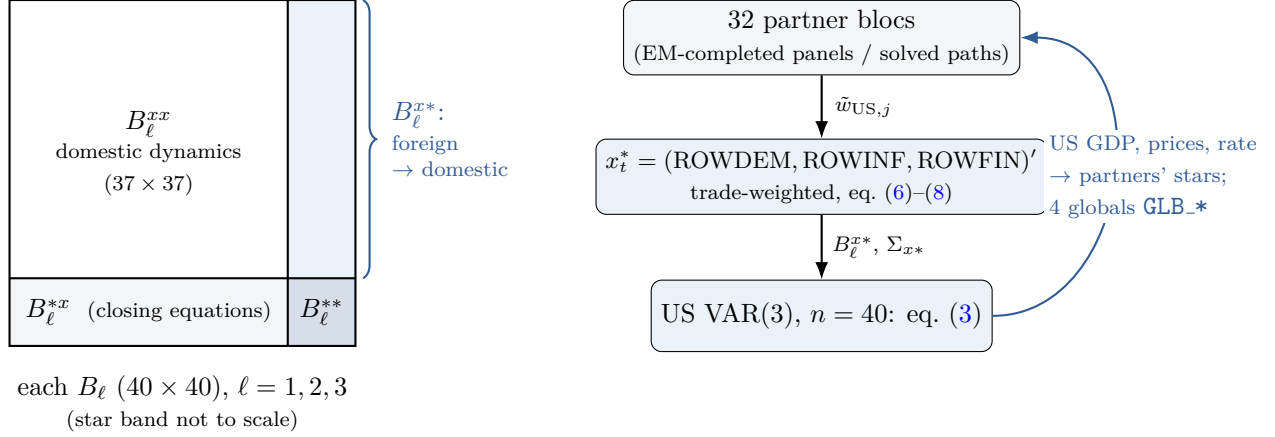


Figure 1: *Left*: where the foreign variables enter — the shaded column block B_ℓ^{**} gives each of the 37 domestic equations its nine foreign regressors; the bottom rows are the stars’ own (closing) equations. *Right*: the two-way linkage. The US is both a receiver (its stars aggregate its partners) and the source of the four global commons; the feedback loop is closed at estimation time by the EM star iteration and at solve time by the fixed point of Section 7.

2. **Contemporaneous transmission** through Σ_{x*} : star innovations are correlated with domestic innovations (e.g. $\text{corr}(u_{\text{GDP}}, u_{\text{ROWDEM}}) = 0.49$, Table 2), and every conditional forecast exploits this covariance through the Kalman gain — pinning a star path moves the domestic variables *within* the same quarter, not only with a lag.
3. **Closing equations** B_ℓ^{*x}, B_ℓ^{**} : the stars have their own three equations, so the closed system (1) can forecast them — this is what replaces the weak-exogeneity assumption of the classical GVAR (Section 8). In single-country (uncoupled) use the star paths are simply forecast by (1); in a coupled solve they are overridden by conditioning (Section 7).

Unlike the country models of Pesaran et al. (2004) and Dees et al. (2007), there is no contemporaneous term $\Lambda_0 x_t^*$ on the right-hand side and no weak-exogeneity restriction: x_t^* is a full endogenous block. The cross-country consistency that the classical GVAR imposes by stacking link matrices is imposed here twice — at estimation time by iterating the star *data* to a fixed point (Section 5), and at solve time by conditioning the star *paths* on the partner blocs’ solutions (Section 7).

2.3 Estimated objects and state-space form

Estimation is by the regression layout $y_t' = z_t' \beta + u_t'$ with $z_t = (1, y_{t-1}', y_{t-2}', y_{t-3}')' \in \mathbb{R}^{121}$, so $\beta = (c, B_1, B_2, B_3)' \in \mathbb{R}^{121 \times 40}$: the first row is the intercept, and rows 2, ..., 121 stack B_1', B_2', B_3' .² This layout (`seasonalRows = 0`) is exact for the US and every seasonally-adjusted bloc; a minority of coupled blocs whose source series are not seasonally adjusted instead carry three quarterly seasonal-dummy coefficients *between* the intercept and the lag blocks, so their $\hat{\beta}$ is $[c; \gamma_{Q2}; \gamma_{Q3}; \gamma_{Q4}; B_1; \dots; B_p]$

²`bvarGLPmf.m:171--183`; the shipped posterior means are $\hat{\beta}$ (`betahat`, 121×40) and $\hat{\Sigma}$ (`sigmahat`, 40×40) in `companion_mean.json`, together with all 500 retained posterior draws in `companion_draws.json.gz` (shapes $[121, 40, 500]$) from which the tool’s predictive bands are computed.

with `seasonalRows = 3` — which is why those blocs’ `companion_mean.json` ships extra rows — and a few coupon-lumpy fiscal blocs additionally have their interest payments evenly distributed within each calendar year before estimation, so their EFFI is an accrual borrowing cost rather than a coupon sawtooth. Neither device touches the US, whose panel is seasonally adjusted and whose interest is not coupon-lumpy.³ For forecasting and conditioning the tool puts (1) in companion (state-space) form,

$$\alpha_t = \mathbf{T} \alpha_{t-1} + c_\alpha + \eta_t, \quad y_t^{\text{obs}} = Z \alpha_t + \varepsilon_t, \quad (4)$$

where $\alpha_t = (y'_t, y'_{t-1}, y'_{t-2})' \in \mathbb{R}^{120}$,

$$\mathbf{T} = \begin{pmatrix} B_1 & B_2 & B_3 \\ I_{40} & 0 & 0 \\ 0 & I_{40} & 0 \end{pmatrix}, \quad c_\alpha = \begin{pmatrix} c \\ 0 \\ 0 \end{pmatrix}, \quad Z = \begin{pmatrix} I_{40} & 0 & 0 \end{pmatrix},$$

$\eta_t \sim \mathcal{N}(0, Q)$ with Σ in the top-left 40×40 block of Q , and $\varepsilon_t \sim \mathcal{N}(0, 10^{-12}I)$ — the near-degenerate measurement noise that implements *hard* conditioning: any observed or pinned cell is matched essentially exactly, and missing cells are simply dropped from the observation vector at that date.⁴

3 The foreign variables

3.1 Trade weights

The linkage weights are bilateral goods-trade shares (exports FOB plus imports CIF, IMF International Merchandise Trade Statistics, 2021–2023 average), row-normalized over the other 32 coupled blocs with a zero diagonal.⁵ The US row has 31 non-zero partners; Table 1 shows the ten largest.

Table 1: The US trade-weight row $w_{\text{US},j}$ (top 10 of 31 partners; shipped values).

Partner	$w_{\text{US},j}$	Partner	$w_{\text{US},j}$
Euro area (EA)	0.176351	Korea (KR)	0.039860
Mexico (MX)	0.168428	United Kingdom (UK)	0.029843
Canada (CA)	0.168175	Taiwan POC (TW)	0.028405
China (CN)	0.144603	India (IN)	0.027841
Japan (JP)	0.049779	Switzerland (CH)	0.019897

³Seasonal-dummy layout and the `seasonalRows` tag: `export_bvar_gvar24_loop.m:331--355` (`nsr = rows($\hat{\beta}$) - (1 + np), asserted $\in \{0, 3\}$; seasonalRows, qoyNext exported to spec.json and companion_mean.json) and app/src/lib/compute/kalman.ts:20--27, 62--71. Even coupon distribution: EVEN_INTEREST={CL,CO,CR,ID,IL,MX,TR,ZA} and even_distribute_years in pipeline/build_fiscal22_columns.py:58--61, 181. Both are described in the seasonality section of the companion user manual.`

⁴`app/src/lib/compute/kalman.ts:43--83` (state space, $R = 10^{-12}$), `kalman.ts:129--141` (missing cells dropped; a finite user-supplied variance overrides the hard pin for *soft* conditioning).

⁵`app/public/data/bvar/gvar/weights_full.json` (the estimator’s input) and `weights_24bloc.json` (the app runtime’s input) — numerically identical, 33 blocs including the US row; builder `pipeline/build_us_weights_row.py` (partners below 0.05% dropped, then renormalized).

The weights are static (one row per bloc), but the weights actually applied are renormalized *per component and per quarter* over the partners available and eligible at that date:

$$\tilde{w}_{\text{US},j,t}^k = \frac{w_{\text{US},j} \mathbf{1}\{j \text{ eligible for component } k \text{ at } t\}}{\sum_{j'} w_{\text{US},j'} \mathbf{1}\{j' \text{ eligible for } k \text{ at } t\}}, \quad k \in \{\text{DEM}, \text{INF}, \text{FIN}\}, \quad (5)$$

so that missing partners' weight is reabsorbed by the rest. Eligibility differs by component: a partner with no rate series at all (Switzerland, Norway, Brazil, Saudi Arabia) drops out of FIN only, and Argentina — whose 2023–24 hyperinflation would otherwise dominate the price and rate moments — is excluded from INF and FIN but kept in DEM.⁶

3.2 Construction of the stars

For each partner j let $\ell_{j,t}^g$ and $\ell_{j,t}^\pi$ be the 100 ln levels of j 's real GDP and consumer-price (or deflator) series, and $r_{j,t}$ its short rate in percent — the bloc's policy rate where one exists, else its 10-year government-bond yield (the fallback used for the Czech Republic, Hungary, and Poland) — all read from j 's *EM-completed* panel (the posterior-mean Durbin–Koopman completion of Section 5, not the raw data). The three US stars are

$$\text{ROWDEM}_t = 4 \sum_{j \neq \text{US}} \tilde{w}_{\text{US},j,t}^{\text{DEM}} \left(\ell_{j,t}^g - \ell_{j,t-1}^g \right), \quad (6)$$

$$\text{ROWINF}_t = 4 \sum_{j \neq \text{US}} \tilde{w}_{\text{US},j,t}^{\text{INF}} \left(\ell_{j,t}^\pi - \ell_{j,t-1}^\pi \right), \quad (7)$$

$$\text{ROWFIN}_t = \sum_{j \neq \text{US}} \tilde{w}_{\text{US},j,t}^{\text{FIN}} r_{j,t}. \quad (8)$$

ROWDEM and ROWINF are trade-weighted quarter-on-quarter *annualized* growth rates in percent — the factor is 4 on the 100 ln scale, i.e. $400 \Delta \ln$, matching the tool's display convention — while ROWFIN is a *level* average of partner rates (no differencing, no annualization).⁷ Partner GDP enters (6) as a chain-linked volume level, with one legacy exception: the estimator's China branch deflates in 100 ln space ($\ell_{\text{CN}}^g - \ell_{\text{CN}}^\pi$), a convention retained from the vintage in which China's panel carried nominal GDP.⁸

Two conventions matter for estimation. First, over the COVID window the two macro stars ROWDEM and ROWINF are set to missing along with the domestic macro block, while ROWFIN

⁶Estimator: `export_bvar_gvar24_loop.m:189--217` (buildstars; the `xinf` exclusion at line 203). Runtime: `app/src/lib/compute/gvarSolve.ts:99--100, 244--273` (`EXCLUDE_INF_FIN={AR}`, `gvarPartnerWeights`). The estimator applies (5) with per-quarter availability; the runtime applies the same per-component renormalization with a static eligibility set — per-quarter availability only matters for the historical star construction.

⁷`export_bvar_gvar24_loop.m:204--212`: `gd(t)=4*ad/wd, gi(t)=4*ai/wi, fr(t)=af/wf`. Leading missing values (quarters before a partner set exists) are back-filled with the first valid value so no bloc carries a NaN prefix (lines 213–217).

⁸`export_bvar_gvar24_loop.m:184--185` (`panelfrom`). The current China raw panel ships a chain-linked volume index, so this branch is a reconciliation item for the next full re-estimation; the app runtime reads the (real) level directly.

stays observed — partners’ policy rates were genuinely observed through 2020 (Section 5). Second, the star *history* is model-generated data: it is rebuilt at every outer sweep from the partners’ current completed panels, which is what makes the estimated system a fixed point rather than a two-step plug-in.

3.3 The global commons

Four of the 37 domestic columns are US-originated variables that every other bloc carries as common exogenous drivers: the 10-year Treasury yield (GLB_USLR), Moody’s Baa yield (GLB_USBAA), a US equity price index (GLB_USEQ), and the WTI oil price (GLB_OIL). In the US bloc they are ordinary endogenous columns of x_t ; in every partner bloc they appear as the same four series and are pinned, at solve time, to the US solution (Section 7). The US is thus the *source* of the global commons and never a receiver.⁹

4 The prior

The prior is the conjugate hierarchical Minnesota–normal–inverse–Wishart of Giannone et al. (2015) (GLP), with the sum-of-coefficients extension of Doan et al. (1984):

$$\text{vec}(\beta) \mid \Sigma \sim \mathcal{N}(\text{vec}(b), \Sigma \otimes \Omega), \quad \Sigma \sim \mathcal{IW}(\Psi, d), \quad d = n + 2 = 42. \quad (9)$$

Prior mean. b is zero except on the first own lag:

$$b_{jj}^{(1)} = \delta_j, \quad \delta_j = \begin{cases} 0 & j \in \{\text{FREV_GDP}, \text{FPEXP_GDP}, \text{SFA}\} \quad (\text{white-noise prior}), \\ 1 & \text{otherwise} \quad (\text{random-walk prior}), \end{cases} \quad (10)$$

i.e. the three fiscal GDP-ratios are shrunk toward stationarity around their means while all other columns — including the three stars — carry the standard random-walk mean.¹⁰

Prior variance. Ω is diagonal. The intercept is diffuse, $\Omega_{11} = V_c \approx 10^6$,¹¹ and the coefficient of variable j at lag ℓ has, in every equation,

$$\Omega_{(\ell,j)} = (d - n - 1) \frac{\lambda^2}{\ell^2 \psi_j} = \frac{\lambda^2}{\ell^2 \psi_j}, \quad (11)$$

⁹Spec blocks "global" at columns 23, 25, 27, 36 of `app/public/data/bvar/US/spec.json`; the broadcast is `applyGlobalPins`, `gvarSolve.ts:479--497`, and is skipped for the US itself (lines 523–526).

¹⁰The switch is the `pos` vector zeroing the own-lag diagonal, `bvarGLPmf.m:204--207`; the tags come from the per-variable `prior` field of the panel spec (`export_bvar_gvar24_loop.m:55--60`). The effective interest rate `EFFI` stays RW.

¹¹ $V_c = 10^7$ in the marginal-likelihood evaluation (`bvarGLPmf.m:33`) and 10^6 in the posterior-draw function (`Support/BVAR.m:3`); both are diffuse and the discrepancy is numerically immaterial, but this note reports what the code does.

since $d - n - 1 = 1$: overall tightness λ , harmonic lag decay with exponent 2, and scaling by ψ_j , the residual variance of a univariate AR(1) fitted to column j . The inverse-Wishart scale is $\Psi = (d - n - 1) \text{diag}(\psi) = \text{diag}(\psi)$, so $\mathbb{E}[\Sigma] = \text{diag}(\psi)$. The ψ_j are *fixed* at their AR(1) estimates (they are not sampled), and no dummy-initial-observation prior is used.¹²

Sum-of-coefficients dummies. The “no-cointegration” (noc) prior is implemented by one artificial observation per variable, with weight $1/\mu$,

$$Y_d = \text{diag}(\bar{y}_0)/\mu, \quad X_d = [0, \text{diag}(\bar{y}_0)/\mu, \text{diag}(\bar{y}_0)/\mu, \text{diag}(\bar{y}_0)/\mu],$$

where $\bar{y}_0$ is the mean of the first p rows of the *regression* sample (observations $p+1, \dots, 2p$ of the panel; 1999Q2–1999Q4). For the three white-noise variables the Y_d rows are zeroed while X_d keeps $\text{diag}(\bar{y}_0)/\mu$, so their dummy observation pulls each equation’s *total* lag loading on them toward zero rather than one — consistent with the $\delta_j = 0$ prior mean, not an exemption from the dummies.¹³

Hyperpriors. The two shrinkage hyperparameters are themselves estimated:

$$\lambda \sim \Gamma(\text{mode } 0.2, \text{sd } 0.4), \quad \mu \sim \Gamma(\text{mode } 1, \text{sd } 1), \quad (12)$$

with support truncated to $[10^{-4}, 5]$, and are sampled by a random-walk Metropolis step on the marginal likelihood of the *completed* data (Section 5). The proposal covariance is the inverse Hessian of the `csmnwel` hyperparameter-posterior maximization with its eigenvalues capped, and the chain starts from a draw around that mode reflected into the support box.¹⁴

5 Estimation with incomplete data

5.1 The missing-data pattern

Interior holes — the pandemic quarters, and interior gaps such as a handful of missing `SFA` cells — are treated as missing data, never as dummies. Two boundary conventions apply: leading-edge gaps (including the first row of the constructed stars) are back-filled with the first valid value, and the likelihood sample is cut at the last fully-observed row of the completed panel — a no-op for the US, whose panel is complete over 1998Q3–2025Q4. The COVID window is fixed at 2020Q2–2020Q4 and masks *macro* cells only: all domestic columns with `isFinancial` = 0 plus the two macro stars `ROWDEM`, `ROWINF` are set to missing there, while financial columns (rates, yields, equity, house prices, credit) and `ROWFIN` remain observed.¹⁵ Because imputation is per cell, the observed

¹²`MNpsi=0, sur=0` in the estimator call, `export.bvar.gvar24.loop.m:107,141`; the Ω and Ψ formulas are `Support/BVAR.m:29--57` and `logMLMFVAR.m:38--63`.

¹³`Support/BVAR.m:20--22, 62--66`; `ydnoc(pos,pos)=0` zeroes only the left-hand side.

¹⁴`cgl_bvarV5.m:31--37, 66--84, 108--117, 151--166`.

¹⁵`export.bvar.gvar24.loop.m:83--90, 130--137`. This differs from the outlier-variance treatment of [Lenza and Primiceri \(2022\)](#): here the pandemic macro observations are simply *not data*, and their imputation is conditioned on the financial columns that are.

financials discipline the imputed macros within the same quarter.

5.2 The Gibbs sampler

Estimation is a data-augmentation sampler in the spirit of [Tanner and Wong \(1987\)](#), run for 1,000 sweeps (500 burn-in). Each sweep g :

1. *Completion draw.* Draw the missing cells with the Durbin–Koopman simulation smoother ([Durbin and Koopman, 2002](#)) on the companion form (4): simulate an artificial path, smooth the difference, and add back, yielding a completed panel $\tilde{y}^{(g)}$. One implementation detail matters for exactness: the smoother’s companion matrices are built *once*, from the `csmnwel` posterior-mode fit ($\hat{\beta}_{\text{mode}}, \hat{\Sigma}_{\text{mode}}$), and are not updated across sweeps — only the artificial path’s innovations use the current Σ draw. The imputed cells are therefore draws from $p(y^{\text{miss}} \mid y^{\text{obs}}, \hat{\beta}_{\text{mode}}, \hat{\Sigma}_{\text{mode}})$, a *fixed-companion approximation* to full data augmentation rather than conditioning on the previous sweep’s coefficient draw.¹⁶
2. *Hyperparameter step.* Update (λ, μ) by random-walk Metropolis on the marginal likelihood $p(\tilde{y}^{(g)} \mid \lambda, \mu)$ of the completed data under (9)–(12).
3. *Coefficient step.* Draw $(\beta, \Sigma)^{(g)}$ from the conjugate normal–inverse-Wishart posterior given $\tilde{y}^{(g)}$ and $(\lambda, \mu)^{(g)}$, including the sum-of-coefficients dummies.

The shipped point model is the posterior mean over the 500 retained draws — $\hat{\beta} = \overline{\beta^{(g)}}$, $\hat{\Sigma} = \overline{\Sigma^{(g)}}$ — and the 500 individual draws ship alongside it for the predictive bands.¹⁷

5.3 The outer star iteration

The star columns (6)–(8) are inputs to the US estimation but outputs of the partners’ estimations — so the whole system is iterated to a fixed point. Each outer sweep m (a *Jacobi* sweep: all 33 blocs are re-estimated in parallel against the stars built from the sweep- $(m-1)$ completions):

$$\text{completed panels}^{(m-1)} \xrightarrow{\text{(6)–(8)}} \text{stars}^{(m)} \xrightarrow{\text{Gibbs above}} \text{completed panels}^{(m)},$$

with convergence declared when

$$\max_{c, k, t} |\text{star}_{c, k, t}^{(m)} - \text{star}_{c, k, t}^{(m-1)}| < 0.03 \quad (13)$$

(annualized percentage points), the COVID rows of the two macro stars excluded from the metric because their sweep-to-sweep churn is imputation Monte-Carlo noise; the cap is 12 sweeps.¹⁸ At

¹⁶`cg1_bvarV5.m:85--95` (companion built from `r.postmax` before the loop and never reassigned inside it) and `:127--145` (the DK draw). Steps 2–3 condition on this completion exactly.

¹⁷`cg1_bvarV5.m:127--205` (the three steps), `export_bvar_gvar24_loop.m:229--247` (export).

¹⁸`export_bvar_gvar24_loop.m:122--157`. The US enters this loop as an ordinary bloc — `US_ENDOGENOUS=1` — after one closed no-stars estimation that seeds the first sweep; that warm-start fit is discarded, not shipped (lines 95–117).

the fixed point every bloc's stars are consistent with every other bloc's completed history — the estimation-time counterpart of the solve-time fixed point below.

6 The estimated foreign block

Table 2 reports the posterior-mean loadings of four representative US equations on the three stars — selected rows of B_ℓ^{x*} in (3) — together with the innovation correlations from $\hat{\Sigma}$. Units matter when reading the numbers: log variables (GDP, exports) are 100 ln *levels*, so a coefficient of 0.042 on lagged ROWDEM means that one percentage point of additional annualized foreign growth for one quarter raises the US GDP level by 0.042 percent (about 0.17 percentage points of annualized growth in that quarter); rate equations (federal funds, 10-year) are in percent per annum.

Table 2: Posterior-mean star loadings of selected US equations, and innovation correlations. Re-computed from the shipped `companion_mean.json`.

Equation	Star regressor	Lag 1	Lag 2	Lag 3	$\sum_\ell$
Real GDP (GDPH)	ROWDEM	+0.042	−0.006	−0.012	+0.024
	ROWINF	+0.053	+0.004	−0.001	+0.056
	ROWFIN	−0.462	+0.308	−0.016	−0.170
CPI (UI)	ROWDEM	+0.026	+0.004	−0.024	+0.007
	ROWINF	+0.072	+0.010	+0.012	+0.095
	ROWFIN	+0.110	−0.064	−0.083	−0.036
Federal funds (POLRATE)	ROWDEM	+0.014	−0.011	−0.003	−0.000
	ROWINF	+0.021	−0.026	+0.012	+0.007
	ROWFIN	−0.086	+0.009	+0.045	−0.032
Real exports (XH)	ROWDEM	+0.072	+0.098	+0.044	+0.214
	ROWINF	−0.135	−0.048	−0.075	−0.258
	ROWFIN	−0.400	−0.010	−0.318	−0.729
<i>Innovation correlations (from $\hat{\Sigma}$):</i> $\text{corr}(u_{\text{GDPH}}, u_{\text{ROWDEM}}) = 0.49$; $\text{corr}(u_{\text{UI}}, u_{\text{ROWINF}}) = 0.54$; $\text{corr}(u_{\text{JGDP}}, u_{\text{ROWINF}}) = 0.43$; $\text{corr}(u_{\text{POLRATE}}, u_{\text{ROWFIN}}) = 0.27$; $\text{corr}(u_{\text{XH}}, u_{\text{ROWDEM}}) = 0.18$.					
<i>Star own-lag persistence (rows of B_ℓ^{**}):</i> ROWDEM: 0.64, −0.11, −0.15 ($\sum = 0.38$); ROWINF: 0.05, −0.19, +0.01 ($\sum = -0.14$); ROWFIN: 0.66, +0.01, −0.05 ($\sum = 0.62$).					

The pattern is economically coherent: exports load positively and persistently on foreign demand ($\sum_\ell = 0.21$) and negatively on foreign rates; the same-quarter co-movement of US GDP with foreign demand (0.49) and of US consumer prices with foreign inflation (0.54) runs through Σ_{x*} , which is precisely the covariance the conditional-forecast smoother exploits when a star path is pinned. Among the star equations themselves, the foreign rate is the persistent one (own-lag sum 0.62), while trade-weighted foreign inflation is close to white noise in its annualized quarterly form.

7 The linkage at solve time

7.1 Scenarios are conditional forecasts

A forecast in the tool is the smoothed mean of (4); a scenario adds *pins* — future observations $y_{i,T+h}^{\text{obs}}$ placed on any cell — and the Durbin–Koopman smoother moves every free variable to its conditional expectation given all pins, exactly in the tradition of conditional VAR forecasting (Waggoner and Zha, 1999; Bańbura et al., 2015). Hard pins carry the 10^{-12} measurement variance of Section 2.3; soft pins (e.g. external forecasts with a trust dial) replace it with a finite variance.

7.2 Pinning the stars to the partners

In a coupled (“global”) solve the US star paths are not left to the closing equations: for every projection step $h = 1, \dots, 20$ they are hard-pinned to

$$x_{\text{US},k,T+h}^{*,\text{pin}} = \bar{x}_{\text{US},k,T+h}^* + \Delta_{\text{US},k,T+h}^*, \quad (14)$$

the bloc’s own *unconditional* star forecast $\bar{x}^*$ plus a trade-weighted sum of the partners’ *deviations* from their own unconditional baselines:

$$\Delta_{\text{US},\text{DEM},T+h}^* = \sum_j \tilde{w}_{\text{US},j}^{\text{DEM}} 4 \left[(\ell_{j,T+h}^{g,\text{scen}} - \ell_{j,T+h-1}^{g,\text{scen}}) - (\bar{\ell}_{j,T+h}^g - \bar{\ell}_{j,T+h-1}^g) \right], \quad (15)$$

$$\Delta_{\text{US},\text{FIN},T+h}^* = \sum_j \tilde{w}_{\text{US},j}^{\text{FIN}} (r_{j,T+h}^{\text{scen}} - \bar{r}_{j,T+h}), \quad (16)$$

and Δ_{INF}^* as (15) on the price levels. At $h = 1$ both back-differences are anchored at the shared panel-origin row (2026Q1 — observed for the US, a cross-bloc nowcast for the other blocs), so the bracketed term in (15) collapses to the level gap $\ell_{j,T+1}^{g,\text{scen}} - \bar{\ell}_{j,T+1}^g$. Deviations — not levels — are what travel between blocs: each bloc keeps its own estimated baseline, and only news relative to baseline propagates.¹⁹ Symmetrically, every partner bloc c has its own ROW^*_c stars pinned the same way — with the US as one of its partners — and its four global commons pinned to the US deviation:

$$z_{c,g,T+h}^{\text{pin}} = \bar{z}_{c,g,T+h} + (z_{\text{US},g,T+h}^{\text{scen}} - \bar{z}_{\text{US},g,T+h}), \quad g \in \{\text{GLB_USLR}, \text{GLB_USBAA}, \text{GLB_USEQ}, \text{GLB_OIL}\}. \quad (17)$$

7.3 The linear fixed point

Because $(\hat{\beta}, \hat{\Sigma})$ are fixed at solve time, the smoothed conditional forecast is *affine* in the pinned values — so the mutual consistency of all 33 blocs is a linear problem. Stack the unknown star

¹⁹`gvarSolve.ts:424--453 (rowPinDelta)` and `499--528 (buildCondPaths)`; the readout column per component mirrors the estimator’s series choice (GDP, CPI/deflator, policy rate), `gvarSolve.ts:106--124, 239`.

deviations of every coupled bloc and the US global deviations into

$$x = \left(\{\Delta_{c,k,T+h}^*\}_{c=1..33, k=1..3, h=1..20}, \{\Delta_{g,T+h}^{\text{GLB}}\}_{g=1..4, h=1..20} \right) \in \mathbb{R}^{2060},$$

($33 \times 3 \times 20 + 4 \times 20 = 2060$). Propagating any candidate x through every bloc’s smoother and re-reading the partners’ responses is an affine map, giving the fixed-point equation

$$x = Mx + d, \tag{18}$$

where d collects the exogenous forcing (the user’s own pins in each bloc, mapped to the deviations they cause), and M — built column-by-column from unit-pin smoother kernels — has spectral radius

$$\rho(M) = 0.9466 < 1, \tag{19}$$

so the solution $x = (I - M)^{-1}d$ exists, is unique, and equals the limit of iterating the blocs to mutual consistency.²⁰ The 2060×2060 inverse is precomputed once and shipped with the app, keyed to the *empty* pin structure: it serves the unconditioned global solve — and drift-only scenarios, which move only d — as a single matrix–vector product. Any hard-pinned scenario changes the pin structure, re-keys M , and is solved by the inverse-free Neumann iteration $x \leftarrow d + Mx$ on a freshly assembled (kernel-cached) M ; soft pins force a Gauss–Seidel sweep over the blocs, because the closed-form kernels assume hard pins.²¹ After solving (18), each bloc runs one final conditional forecast with its stars and globals pinned to (14)–(17), so every displayed variable is consistent with the fixed point. The blocs are aligned by *projection step*, not calendar date: the blocs share the uniform 2026Q1 origin, so step h is the same quarter everywhere — with one documented exception, Israel, whose rows sit on its own calendar (origin 2025Q1) and which is coupled by row with a four-quarter offset pending its data re-pull.²²

7.4 What endogeneity buys

Since the US participates in x with its own sixty star-deviation entries, a shock anywhere in the system feeds back to the United States: a scenario that depresses euro-area demand lowers ROWDEM_{US} through (15), moves US GDP through B_ℓ^{x*} and Σ_{x*} , and the US response then re-enters every partner’s stars and the four global commons — the Rest-of-World→US channel that a frozen weakly-exogenous anchor cannot represent. With the legacy frozen-anchor configuration the

²⁰Computed by direct eigendecomposition of the shipped M ; the dominant eigenvalues are the complex pair $0.9446 \pm 0.0305i$. Before 2026-07-17 the artifact’s `rhoSpectral` field read 0.935044, produced by a power iteration on M^2 that does not converge on a complex dominant pair; the estimator and the shipped field (now 0.945494) were corrected on that date. Everything that matters analytically is that $\rho(M) < 1$.

²¹`gvarSolve.ts:1112--1494` (`buildLinearSystemEndo`, `solveGVARClosedFormEndo`); artifact `app/public/data/bvar/gvar/coupling-default.json.gz` (dim: 2060, pinStructure: empty, M and $(I - M)^{-1}$ at 6 significant digits); producer `app/scripts/precompute_gvar_coupling.mjs`; routing `runGvarSolve.ts:229--253`. Affinity of the smoother in the pin values for a fixed pin mask: `kalman.ts:479--504`.

²²`app/src/lib/dates.ts:57--69`.

same architecture degrades gracefully: the US drops out of x , and the system solves with one-way US→world transmission only.

8 Relation to the classical GVAR

Table 3 summarizes the mapping to [Pesaran et al. \(2004\)](#), [Dees et al. \(2007\)](#), and the survey treatment of [Chudik and Pesaran \(2016\)](#).

Table 3: The US bloc versus the classical GVAR country model.

	Classical GVAR (VARX*)	MFFM US bloc
Foreign variables	x_t^* on the right-hand side with contemporaneous and lagged terms ($\Lambda_0 x_t^* + \Lambda_1 x_{t-1}^*$), assumed weakly exogenous (tested per bloc)	x_t^* endogenous columns with their own equations; linkage through lagged B_ℓ^{x*} and the innovation covariance Σ_{x*} ; no weak-exogeneity assumption for the US
Estimation	Equation-by-equation OLS / reduced-rank ML, each bloc on its own sample	Hierarchical Bayes (GLP) with missing-data completion, iterated with the partners to the star fixed point (13)
Missing data	Common solve window; ragged edges truncated	2020Q2–Q4 macro cells and interior gaps imputed by the DK smoother inside estimation; ragged panel starts kept
Dominant unit	US model restricted; US weakly exogenous to the rest	US fully endogenous, with its own stars and a trade-weight row; RoW→US feedback
Global solve	Stack link matrices, invert G_0 once: $x_t = G_0^{-1}(a + \sum_\ell G_\ell x_{t-\ell} + u_t)$	Deviation fixed point $x = Mx + d$ in pin space, $\rho(M) = 0.947$; pre-shipped $(I - M)^{-1}$; scenario-time conditioning via the DK smoother

9 The model in numbers

Table 4: The shipped US bloc at a glance.

Endogenous variables n	40 (37 domestic + 3 stars)
Lags p / horizon H	3 / 20 quarters
Estimation sample	1998Q3–2025Q4 ($T = 110$); display origin 2026Q1
Forecast window shown	2026Q2–2031Q1
COVID missing window	2020Q2–2020Q4, macro cells only
Posterior draws	1,000 Gibbs sweeps, 500 retained (shipped)
Prior	GLP Minnesota–NIW; $\lambda \sim \Gamma(0.2, 0.4)$, $\mu \sim \Gamma(1, 1)$ (mode, sd)
White-noise columns	FREV_GDP, FPEXP_GDP, SFA
Trade weights	IMF IMTS 2021–23 average; 31 partners; EA 0.176, MX 0.168, CA 0.168, CN 0.145
Global commons sourced	GLB_USLR, GLB_USBAA, GLB_USEQ, GLB_OIL
Coupled system	33 blocs; state dim 2060; $\rho(M) = 0.945494$ (eigendecomposition-exact; see the footnote to eq. (19)); $(I - M)^{-1}$ pre-shipped
Hard-pin measurement variance	10^{-12}

A The 40-variable roster

Order, mnemonics, spaces, and priors as shipped in `app/public/data/bvar/US/spec.json`. Descriptions are from the estimation panel `DataRaw.US_glob.json`, and the source mnemonics are from the shipped `coverage_manifest.json`. *Space*: 100 ln = enters as 100 ln(level), displayed as 4Δ annualized growth (five index-type series display as levels); level = enters untransformed (percent, percent of GDP, or index points). *Prior*: RW = random-walk own-lag mean 1; WN = white-noise own-lag mean 0, eq. (10).

Table 5: The US variable roster, in shipped order.

#	Mnemonic	Description	Space	Prior	Block	Source
0	GDPH	Real Gross Domestic Product	100 ln	RW	macro	FRED GDPC1
1	CH	Real Personal Consumption Expenditures	100 ln	RW	macro	FRED PCECC96
2	FRH	Real Private Residential Fixed Investment (nominal, deflated)	100 ln	RW	macro	FRED PRFI / GDPDEF
3	FNH	Real Private Nonresidential Fixed Investment (nominal, deflated)	100 ln	RW	macro	FRED PNFI / GDPDEF
4	XH	Real Exports of Goods & Services	100 ln	RW	macro	FRED EXPGSC1
5	MH	Real Imports of Goods & Services	100 ln	RW	macro	FRED IMPGSC1
6	GH	Real Government Consumption & Gross Investment	100 ln	RW	macro	FRED GCEC1
7	JGDP	GDP Chain Price Index	100 ln	RW	macro	FRED GDPDEF
8	JC	PCE Chain Price Index	100 ln	RW	macro	FRED PCEPI
9	JCXFE	PCE less Food & Energy Price Index	100 ln	RW	macro	FRED PCEPILFE
10	UI	CPI-U: All Items	100 ln	RW	macro	FRED CPIAUCSL
11	UIXFDG	CPI-U: All Items Less Food & Energy	100 ln	RW	macro	FRED CPILFESL
12	LXBC	Business Sector Compensation Per Hour (nominal)	100 ln	RW	macro	FRED HCOMPBS
13	LANAGRA	All Employees: Total Nonfarm	100 ln	RW	macro	FRED PAYEMS
14	LR	Civilian Unemployment Rate	level	RW	macro	FRED UNRATE
15	IP	Industrial Production Index	100 ln	RW	macro	FRED INDPRO
16	CUMFG	Capacity Utilization: Manufacturing	level	RW	macro	FRED MCUMFN
17	HST	Housing Starts	100 ln	RW	macro	FRED HOUST
18	YPDH	Real Disposable Personal Income	100 ln	RW	macro	FRED DPIC96
19	CSENT	U. Michigan Consumer Sentiment	level	RW	macro	FRED UMCSENT
20	POLRATE	Effective Federal Funds Rate	level	RW	fin.	FRED FEDFUNDS
21	FCM2	2-Year Treasury Yield (constant maturity)	level	RW	fin.	FRED DGS2
22	FCM5	5-Year Treasury Yield (constant maturity)	level	RW	fin.	FRED DGS5
23	GLB_USLR	10-Year Treasury Yield (constant maturity)	level	RW	global	FRED DGS10

#	Mnemonic	Description	Space	Prior	Block	Source
24	FAAA	Moody's Seasoned Aaa Corporate Yield	level	RW	fin.	FRED AAA
25	GLB_USBAA	Moody's Seasoned Baa Corporate Yield	level	RW	global	FRED BAA
26	FXTWM	Trade-weighted US\$ index (advanced foreign economies)	100 ln	RW	fin.	FRED DTWEX-AFEGS (spliced onto DTWEXM)
27	GLB_USEQ	US equity price index	100 ln	RW	global	FRED SPASTT01USM661N
28	GSNE	IMF non-fuel commodity price index	100 ln	RW	fin.	FRED PNFUELIN-DEXM
29	SP500VOL	CBOE VIX	level	RW	fin.	FRED VIXCLS
30	HP	Case-Shiller national house price index	100 ln	RW	fin.	FRED CSUSHPISA
31	LEV	Real nonfinancial business debt	100 ln	RW	fin.	FRED BCNSDODNS / GDPDEF
32	FREV_GDP	Federal receipts (% of GDP)	level	WN	fiscal	FRED FGRECPT / GDP
33	FPEXP_GDP	Federal primary expenditure (% of GDP)	level	WN	fiscal	FRED FGEXPND, A091RC1Q027SBEA, GDP
34	EFFI	Federal effective interest rate (% p.a.)	level	RW	fiscal	FRED A091RC1Q027SBEA, GFDEGDQ188S
35	SFA	Stock-flow adjustment (% of GDP)	level	WN	fiscal	debt-identity residual
36	GLB_OIL	WTI crude oil price (quarterly average)	100 ln	RW	global	FRED WTISPLC
37	ROWDEM_US	Foreign demand: trade-weighted partner real-GDP growth, eq. (6)	level	RW	star	constructed
38	ROWINF_US	Foreign inflation: trade-weighted partner inflation, eq. (7)	level	RW	star	constructed
39	ROWFIN_US	Foreign rate: trade-weighted partner policy rate, eq. (8)	level	RW	star	constructed

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